THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A train leaves at 6 a.m. and runs at a steady 60 km/h. By 12 noon it has covered: 1

(a) 720 km **(b)** 120 km **(c)** 60 km **(d)** 360 km

2. A person breathes 16 times a minute, taking in 500 mL each time. The volume of air in one minute is: 1

(a) 16 mL **(b)** 8000 mL **(c)** 4000 mL **(d)** 500 mL

3. Which energy change occurs in respiration? 1

(a) heat to chemical **(b)** sound to chemical

(c) light to chemical **(d)** chemical energy to usable energy

4. During a lunar eclipse, the Moon passes into Earth's: 1

(a) shadow **(b)** atmosphere only **(c)** orbit only **(d)** light

5. The Moon takes 27.3 days to complete a revolution but 29.5 days from full moon to full moon because: 1

(a) the Moon slows down **(b)** the Earth moves around the Sun meanwhile

(c) the Sun moves **(d)** the Moon stops

6. Which statement about milk of magnesia is correct? 1

(a) it is a neutral salt used as a drink **(b)** it is an indicator

(c) it is an acid used to relieve acidity **(d)** it is a base used to relieve acidity

7. In the same tester the bulb does not glow when a dry wooden pencil case is used. The case is: 1

(a) a conductor **(b)** an insulator **(c)** a filament **(d)** a battery

8. Which of these metals is most suitable for making a very hot cooking pan? 1

(a) copper **(b)** wood **(c)** lead **(d)** sulphur

9. A farmer wants to keep tools from rusting during the rainy season. Which is best? 1

(a) apply oil and keep dry **(b)** keep in a damp shed

(c) wash daily with salt water **(d)** leave outside

10. In which year is the average growth rate highest for girls? 1

(a) around 40 **(b)** around 11 to 13 years

(c) around 20 **(d)** around 2

11. Which of the following is a correct application of conduction? 1

(a) sunlight **(b)** ceiling fan **(c)** sea breeze **(d)** heating iron rod

12. A sprinter's reaction time is 0.2 s and a stopwatch is started after seeing the start. The measured 10 s may be: 1

(a) double **(b)** less than true by about 0.2 s

(c) more than true **(d)** exact

13. Herbivores have a longer small intestine than carnivores because: 1

(a) they sleep **(b)** they are bigger

(c) they are slower **(d)** plant food takes longer to digest

14. A farmer grows wheat after pulses and applies less urea. The reason is: 1

(a) urea is costly **(b)** wheat needs no nitrogen
(c) pulses consume nitrogen **(d)** nitrogen left by pulses

15. A red rose in white light is viewed in a plane mirror. The image of the rose appears: 1

(a) red **(b)** green **(c)** white **(d)** black

16. A student claims 'the Sun is closer to Earth in summer'. A good test is: 1

(a) ask friends
(b) look at the Sun
(c) compare dates of nearest distance with seasons
(d) measure the temperature once

17. Liquid A turns China rose pink, liquid B green. If A and B are mixed in equal strengths, the mixture most probably: 1

(a) becomes more acidic **(b)** becomes more basic
(c) turns almost neutral **(d)** turns blue

18. A chain of 10 decorative bulbs goes off when one bulb burns out. The bulbs are connected in: 1

(a) open circuit **(b)** parallel **(c)** series **(d)** no circuit

19. A metal Z reacts with dilute HCl giving hydrogen but does not react with cold water. Z may be: 1

(a) sodium **(b)** zinc **(c)** gold **(d)** potassium

20. A solution turns milky when a gas is passed. The gas is probably: 1

(a) oxygen **(b)** carbon dioxide **(c)** nitrogen **(d)** hydrogen

21. A student with tiredness, pale tongue and poor concentration has these symptoms mostly due to: 1

(a) anaemia **(b)** rickets **(c)** scurvy **(d)** goitre

22. A candle is placed under a paper spiral hung from a thread. The spiral rotates because: 1

(a) heat radiation pushes **(b)** wind from the room
(c) hot air rising pushes it **(d)** cold air pushes it

23. A man runs round a circular track of length 400 m in 100 s. His average speed is: 1

(a) 400 m/s **(b)** 4 m/s **(c)** 40 m/s **(d)** 0 m/s

24. A person's stomach acid is removed. Which is the most likely effect? 1

(a) faster starch digestion
(b) poor protein digestion and more germs reaching the intestine
(c) no change
(d) better fat digestion

25. In a set-up with a plant in a bell jar with NaOH solution, starch is absent in the leaf because: 1

(a) NaOH adds oxygen **(b)** NaOH absorbs carbon dioxide
(c) light is blocked **(d)** NaOH gives CO_2

26. The image formed by a plane mirror of a person 1.6 m tall is: 1

(a) 0.8 m tall **(b)** 0.4 m tall **(c)** 3.2 m tall **(d)** 1.6 m tall

27. If the Earth revolved around the Sun in half the time, a year would have about: 1

(a) 365 days **(b)** 730 days **(c)** 91 days **(d)** 182 days

28. A farmer needs 200 kg of lime for 1 hectare of acidic soil. For 2.5 hectares, the lime needed is: 1

(a) 250 kg **(b)** 800 kg **(c)** 400 kg **(d)** 500 kg

29. In the circuit tester, the bulb glows brightly with a copper wire and dimly with graphite. This shows: 1

(a) graphite conducts better **(b)** copper conducts better
(c) both conduct equally **(d)** graphite is an insulator

30. In the reaction above, the colour of the solution changes from: 1

(a) blue to green **(b)** colourless to blue
(c) green to blue **(d)** blue to colourless

31. A student heats a piece of wax and sees it melt. On stronger heating, black smoke appears. The black smoke indicates: 1

(a) melting only **(b)** no change
(c) evaporation only **(d)** unburnt carbon, a chemical change

32. If testes were removed in an adult, secondary sexual characters may: 1

(a) double **(b)** become female
(c) stay unchanged forever **(d)** be affected over time

33. If the sea and the land were heated equally, the breezes would: 1

(a) be reversed **(b)** not form **(c)** be of storms **(d)** be stronger

34. A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

(a) 12 km **(b)** 6 km **(c)** 5 km **(d)** 24 km

35. In one hour this volume is: 1

(a) 8 litres **(b)** 480 litres **(c)** 48 litres **(d)** 4800 litres

36. Which of these is a net result of photosynthesis in a day? 1

(a) less glucose **(b)** no glucose
(c) more glucose than used in respiration **(d)** only oxygen

37. A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

(a) four times **(b)** double **(c)** same **(d)** half

38. The Sun is much larger than the Moon, yet in a total solar eclipse the Moon can hide it because the Moon is: 1

(a) larger **(b)** much hotter
(c) brighter **(d)** much nearer to the Earth

39. A tablet that fizzes in water is likely to contain: 1

(a) only a base **(b)** a carbonate and an acid
(c) only salt **(d)** only sugar

40. A tester glows dimly with a pencil lead and brightly with a copper wire. This shows that: 1

(a) both conduct, copper better **(b)** copper is an insulator
(c) neither conducts **(d)** only graphite conducts

41. **Assertion (A):** Day and night occur because of the rotation of the Earth. 1
Reason (R): The Earth rotates on its axis once in about 24 hours.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Red litmus paper turns blue when dipped in lemon juice. 1
Reason (R): Lemon juice is acidic in nature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** A fuse is connected in series with an electrical appliance. 1
Reason (R): Fuse wire has a very high melting point, so it never melts.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** Graphite conducts electricity even though it is a non-metal. 1
Reason (R): Graphite is a form of carbon.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Rusting of iron takes place only when both oxygen and water are present. 1
Reason (R): Iron is a metal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The pituitary gland is called the master gland. 1
Reason (R): The pituitary gland secretes the hormone insulin.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The heat of the Sun reaches the Earth by radiation. 1
Reason (R): Conduction and convection need a material medium, which is almost absent between the Sun and the Earth.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The time period of a simple pendulum increases when its length is increased. 1
Reason (R): A longer pendulum takes more time to complete one oscillation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Bile, which is made by the liver, helps in the digestion of fats. 1
Reason (R): Bile contains enzymes that break fats into fatty acids.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Stomata help in the exchange of gases in a leaf. 1
Reason (R): Guard cells control the opening and closing of stomata.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**